

equiripple. The simplest technique used to develop digital FIR filters is windowed filters. This technique is based on designing a filter using well-known frequency-domain transition functions called windows.

FIR can be implemented in various ways but rectangular filters and Hamming filters have proved to be the most practical ones as these provide appreciable outputs compared to the other types. FIR filters can be used in a system that doesn't require much accuracy. The FIR program codes for rectangular (`rectangular.m`) and Hamming filters (`hamming.m`) are given at the end of this article.

Rectangular filter. Fig. 3 shows the magnitude plot for a rectangular filter. The plot gives a large number of side lobes. This can result in a large amount of distortion in the output. Hence another hardware system may be required to enhance the output. The angle plot of the rectangular filter is shown in Fig. 4.

Hamming filter. Hamming filters provide lower side lobes than rectangular filters. Also, rectangular filters have a very sharp transition, which is practically impossible to implement. Some sort of distortion is introduced due to the noise.

The 'raised cosine' with these particular coefficients was proposed by Richard W. Hamming. The window is optimised to minimise the maximum (nearest) side lobe, giving it a height of about one-fifth that of the Hann window – a raised cosine with simpler coefficients. The Hamming window equations are:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \dots\dots\dots \text{Eq. (4)}$$

$$w_0(n) = w\left(n + \frac{N-1}{2}\right) \\ = 0.54 + 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \dots\dots\dots \text{Eq. (5)}$$

where, N = width of the filter sample of discrete time-symmetrical window function

n = integer with values $0 \leq n \leq N-1$

The signal input to Hamming filter is shown in Fig. 5. The `hamming.m` code is written as per equations 4 and 5

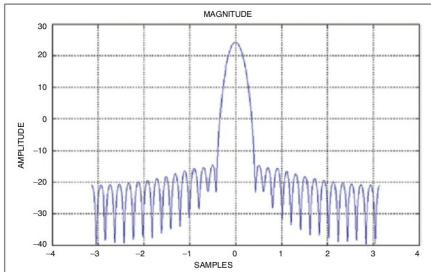


Fig. 6: Magnitude of Hamming filter output

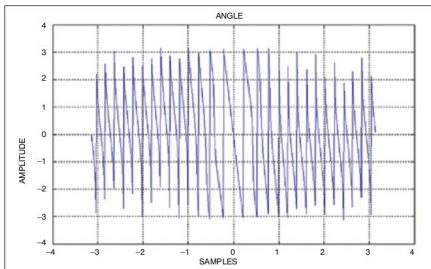


Fig. 7: Angle of Hamming filter output

given above. The program output plot of magnitude and angle for the Hamming filters is shown in Figs 6 and 7, respectively.

It can be observed that the magnitude transition is very smooth for IIR filter. FIR does have a smooth cut-off but it still gives a satisfactory result with the magnitude and phase.

IIR filters are preferred to FIR because:

1. FIR requires more coefficients for sharp cut-off filters than IIR, thus requiring more processing time and storage for FIR implementation.

2. Analogue filters can be readily transformed into equivalent IIR digital

filters at given specifications.

3. IIR filters have a high throughput.

4. FIR is algebraically more difficult to synthesise.

Some MATLAB functions

1. 'Semilogx' and 'semilogy' plot data as logarithmic scales for the x- and y-axis, respectively. 'semilogx(Y)' creates a plot using base 10 logarithmic scale for the x-axis and a linear scale for the y-axis. It plots the columns of Y versus their index if Y contains real numbers. 'semilogx(Y)' is equivalent to 'semilogx(real(Y), imag(Y))' if Y contains complex numbers. 'semilogx'